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YanGo Smart Mobility Platform End-to-End Solution & API Specification

What every portal needs from the backend, how the front end is wired, and what the tracking, mapping and AI engines have to deliver — for the Yangon Region bus network and its highway services.

Version 1.0 — working specification

Pilot tenant Yangon Region Transport Committee (YRTC)

Channels Citizen App · Driver App · Bus Operator Portal · Licensing Portal · Authority Console · Public departure board

Status Derived from the working demo build; endpoint contracts to be ratified with YRTC before implementation

Contents

1. Purpose and how to read this document

2. System context

1. Actors and channels
2. Tenancy and jurisdiction model
3. The one rule that shapes the whole design

3. Service architecture

1. Thirty-four services in eight domains
2. Shared platform components

4. API conventions

1. Base URL, versioning, authentication
2. Errors, pagination, idempotency, rate limits
3. Audit and reason enforcement

5. API specification by domain

1. Identity and access
2. Registry: operators, vehicles, drivers
3. Licensing and permits
4. Network, schedules and GTFS
5. Ticketing, seats and fares
6. Payments and settlement
7. Tracking, telemetry and ETA
8. Boarding and validation
9. Safety, SOS and incidents
10. Compliance and enforcement
11. Content, notifications and advertising
12. Analytics, reporting and open data

6. API scope per portal

7. Front-end specification

1. Screen inventory
2. Navigation map per portal
3. Screen-to-endpoint binding
4. State, caching and offline behaviour

8. Tracking and maps engine

9. AI scope

10. External integrations

11. Non-functional requirements

12. Delivery phases and open questions

1. Purpose and how to read this document

This specification describes what has to be built behind the YanGo Smart Mobility Platform so that the six channels already prototyped can run against real services. It is written for the engineering and architecture teams who will implement the backend, and for the YRTC counterparts who will review scope before contracts are cut.

Three questions are answered, in this order:

1. **What APIs does each feature need?** — Section 5 lists every endpoint, grouped by service domain, with its purpose, its caller and the OAuth scope that guards it.
2. **What does each portal call?** — Section 6 maps the endpoints onto the six channels, so a team building one portal can see its own surface without reading the rest.
3. **How is the front end wired?** — Section 7 gives the screen inventory, the navigation graph per portal and the endpoint each screen binds to.

Sections 8 to 10 cover the parts that are usually underestimated: the tracking and mapping engine, the AI scope, and the external systems this platform has to talk to. Section 11 states the non-functional targets the whole design has to meet.

Reading the endpoint tables. Paths are relative to the API base. A path segment in braces is a path parameter. The *scope* column gives the OAuth 2.0 scope required; an endpoint with no scope is public. Endpoints marked **M** are needed for the pilot, **S** for the first production release, and **C** may be deferred.

2. System context

2.1 Actors and channels

Six channels, five human audiences, one database. Nothing is re-keyed between them: what an operator files in Licensing becomes the permit the Operator Portal plans against, the shift the Driver App can start, the bus the Citizen App can sell a seat on, and the line the Authority Console measures.

Channel	Primary users	Device	Works offline	Authentication
Citizen App	Passengers in Yangon Region	Phone (PWA)	Ticket QR, saved trips	Phone number + 6-digit OTP
Driver App	Bus drivers and conductors	Phone (PWA)	Shift events, scans, SOS queue	Phone + OTP, face match on shift start
Bus Operator Portal	PO owners, ops, finance, safety	Desktop	No	Email + password + mandatory 2FA
Licensing Portal	Applicants and licensing officers	Desktop	No	Email + password + 2FA; officers via government SSO
Authority Console	YRTC officers, ministry, auditors	Desktop	No	Government SSO + 2FA, jurisdiction-scoped
Public departure board	Terminal hall display	Kiosk / large screen	Cached last board	Device token, read-only

2.2 Tenancy and jurisdiction model

Five tiers, and every record carries the tier it belongs to. A regional officer never sees another region's rows; an operator never sees another operator's. This is enforced in the data layer by a jurisdiction predicate applied to every query, not by hiding buttons in the UI.

- **National** — Ministry of Transport: policy, thresholds, national dashboard, audit body.
- **Region** — YRTC and its counterparts in Mandalay and Bago: licensing, compliance, fare bands.
- **Terminal** — terminal managers: bays, gate check-in, crowd, departure board.
- **Operator** — the licensed bus company: its own fleet, drivers, schedules, sales.
- **Field** — drivers and conductors: one shift, one bus, one manifest.

2.3 The one rule that shapes the whole design

A regulator's number must mean the same thing at every level. The load factor a conductor sees, the load factor the operator reports and the load factor the ministry publishes are the same figure computed once, in one service, from one definition. Every KPI in this platform is therefore computed server-side in the analytics service and read by all channels — never recomputed in a client. Where a figure cannot be trusted because too few vehicles were transmitting, the API returns `not_measurable` rather than a number, and the coverage index that led to that verdict travels with the response.

3. Service architecture

3.1 Thirty-four services in eight domains

Services are deployed behind one API gateway. Each owns its data; cross-domain reads go through the owning service's API, never straight into another service's tables.

#	Domain	Service	Responsibility	Consumers
1	Identity	IAM / SSO	OTP, password + 2FA, government SSO, tokens, sessions	All channels
2	Identity	Authorization	RBAC + jurisdiction scope, maker-checker, reason enforcement	All consoles
3	Identity	eKYC	ID OCR, liveness, face match, registry lookup	Licensing, Driver App, Citizen App
4	Identity	Tenant & jurisdiction	National → region → city → terminal → operator	Authority Console, platform admin
5	Registry	Operator registry	Company master, depots, bank accounts	Licensing, Operator Portal, Authority
6	Registry	Vehicle registry	Vehicles, seat layouts, devices, documents	Operator Portal, Licensing, Citizen App
7	Registry	Driver & crew registry	Drivers, operator binding, violations	Driver App, Operator Portal, Licensing
8	Registry	Licensing workflow	Applications, states, inspections, approvals	Licensing Portal, Authority
9	Registry	Permit & e-certificate	Terms, signed PDF/QR, public verification	All portals, public verify page
10	Registry	Document service	Upload, virus scan, OCR, versions, retention	Licensing, Operator Portal

#	Domain	Service	Responsibility	Consumers
11	Network	Network master (GTFS)	Routes, stops, terminals, shapes, calendars	All channels, open data feed
12	Network	Schedule & timetable	Trips, frequencies, permit validation	Operator Portal, Citizen App, Terminal
13	Network	Roster & shift	Assignments, hours-of-service checks	Operator Portal, Driver App
14	Network	Journey planner	Multi-leg routing with walking and transfers	Citizen App, chatbot
15	Sales	Fare & product	Fares, caps, concessions, passes, promos	Citizen App, Operator Portal
16	Sales	Inventory & seat	Seat maps, holds, channel quotas	Citizen App, counter sale, agents
17	Sales	Booking & order	Cart, PNR, passenger records	Citizen App, Operator counter
18	Sales	Ticket & validation	Signed QR, validation, check-in, offline keys	Citizen App, Driver App, terminal gate
19	Sales	Refund & reschedule	Policy evaluation, refunds, reschedules	Citizen App, Operator Portal, helpdesk
20	Money	Payment orchestration	Provider routing, callbacks, idempotency	Citizen App, Licensing, ads billing
21	Money	Wallet & ledger	Double-entry ledger, passenger wallet	Citizen App, finance views
22	Money	Settlement & split	Split rules, T+1 payouts, reconciliation	Operator finance, Authority settlement
23	Money	Government billing	Licence fees, fines, invoices, treasury posting	Licensing, Authority revenue
24	Operations	Telematics ingest	Device registry, GPS/CAN/APC ingest, device health	Real-time engine, Coverage Index
25	Operations	Real-time & ETA	Vehicle state, ETA, GTFS-RT, bunching, Coverage Index	Citizen App, consoles, open data
26	Operations	Trip & event	Trip lifecycle, manifests, no-shows	Driver App, Operator Portal, analytics
27	Operations	Safety & incident	SOS, incidents, SLA, dispatch, evidence	All channels
28	Operations	Compliance & enforcement	Measurements, findings, disputes, sanctions, e-fines	Authority, Operator Portal
29	Engagement	Notification	Multi-channel delivery, templates, preferences	All channels
30	Engagement	Content CMS	Alerts, announcements, FAQs, terminal info	Citizen App, boards, chatbot KB
31	Engagement	Ads	Inventory, campaigns, delivery, billing	Citizen App, in-bus and terminal screens
32	Engagement	Helpdesk & feedback	Complaints, ratings, lost & found	Citizen App, Authority, Operator Portal
33	Data & AI	Analytics & reporting	Data lake, k-anonymous marts, dashboards, report builder	Authority, Operator Portal
34	Data & AI	AI / ML & chatbot	Models, feature store, LLM gateway, retrieval	All channels

3.2 Shared platform components

API gateway Append-only hash-chained audit log Versioned configuration and threshold store Search Job scheduler Object storage

Observability stack (logs, metrics, tracing)

The **append-only, hash-chained audit log** and the **versioned configuration store** are not optional infrastructure. Every regulatory decision in this platform must be reconstructable: who did what, to which object, under which threshold version, with what written reason. A change to a threshold never rewrites findings or reports already issued.

4. API conventions

4.1 Base URL, versioning, authentication

Base URL	https://api.yango.gov.mm/v1
Versioning	major version in the path; additive changes never bump it, removals always do
Transport	TLS 1.3 only; HSTS; no plaintext fallback
Auth	OAuth 2.0 + OIDC. Bearer access token, 15 min; refresh token, 30 days, rotating
Citizen / driver	phone + OTP grant, device-bound refresh token
Console users	authorization-code + PKCE via government SSO, 2FA mandatory
Machine clients	client-credentials; operator API keys are scoped to one operator

Every token carries `sub`, `role`, `jurisdiction` and `operator_id` where relevant. The gateway rejects a request whose jurisdiction claim does not cover the resource before it reaches the service.

4.2 Errors, pagination, idempotency, rate limits

Concern	Contract
Errors	RFC 9457 problem+json: <code>type</code> , <code>title</code> , <code>status</code> , <code>detail</code> , <code>instance</code> , plus <code>errors[]</code> for field-level validation. Never leak internal identifiers or stack traces.
Pagination	Cursor-based: <code>?limit=50&cursor=...</code> ; response carries <code>next_cursor</code> . Offset paging is not offered on any list that can grow unbounded.
Filtering	Explicit query parameters only; no generic query language on public endpoints.
Sorting	<code>?sort=field</code> or <code>-field</code> , whitelisted per endpoint.
Idempotency	Every POST that moves money or issues a document requires an <code>Idempotency-Key</code> header. Keys are stored 24 h; a repeat returns the original response, never a second charge or a second permit.
Concurrency	Optimistic locking with <code>If-Match</code> on <code>ETag</code> for every mutable regulatory record.
Rate limits	Per token and per IP at the gateway. Citizen read 120/min, write 20/min. Operator API 600/min. Telemetry ingest is exempt and shaped separately.
Time	All timestamps RFC 3339 with offset. The platform stores UTC; Asia/Yangon (UTC+06:30) is a presentation concern.
Money	Integer minor units with an explicit <code>currency</code> . MMK has no minor unit, so amounts are whole kyat.
Localisation	<code>Accept-Language</code> : <code>my</code> or <code>en</code> . Names carry both <code>name</code> and <code>name_mm</code> .

4.3 Audit and reason enforcement

A set of endpoints may not be called without a written reason. The reason is a required body field, minimum 20 characters, and it is written to the audit log with the actor, the object, the threshold version in force and the request id. The gateway rejects the call outright if the field is missing.

- Any change to a fare band, threshold, permit term or service standard.

- Any approval, rejection, suspension or revocation.
- Any export of line-level or operator-level figures.
- Any override of an automatic block (a failed inspection, an expired medical, a coverage floor).

Maker-checker. The officer who verified an application may not approve it. The API enforces this server-side: POST `/licensing/applications/{id}/approve` returns 409 `separation_of_duties` when the caller is the same subject that filed the verification.

5. API specification by domain

183 endpoints across twelve domains. Scopes follow the pattern `domain:action`; `*`:`read` is granted to every authenticated console user within their jurisdiction.

5.1 Identity and access

Method	Path	Purpose	Called by	Scope
POST	/auth/otp/request	Send a 6-digit code to a phone number; rate-limited per number and per device	Citizen App, Driver App	—
POST	/auth/otp/verify	Exchange the code for an access and refresh token pair	Citizen App, Driver App	—
POST	/auth/token	Refresh, or client-credentials grant for machine clients	All channels	—
POST	/auth/logout	Revoke the refresh token and the device binding	All channels	—
GET	/auth/sso/authorize	Start the government SSO authorization-code flow with PKCE	Consoles	—
POST	/auth/2fa/enrol	Enrol a TOTP authenticator; mandatory before first console sign-in	Consoles	self:write
POST	/auth/2fa/verify	Second factor on sign-in and on every privileged action	Consoles	self:write
GET	/me	The caller: identity, role, jurisdiction, operator binding, capabilities	All channels	self:read
PATCH	/me	Update own profile, language, accessibility and notification preferences	All channels	self:write
GET	/me/capabilities	The permission matrix the UI uses to hide what the caller may not do	Consoles	self:read
POST	/me/devices	Register a device for push and bind the refresh token to it	Citizen App, Driver App	self:write
DELETE	/me/devices/{id}	Sign a lost device out remotely	Citizen App, Driver App	self:write
GET	/users	List console users in the caller's jurisdiction or operator	Operator Portal, Authority	users:read
POST	/users	Invite a console user; the invitee enrolls 2FA before first sign-in	Operator Portal, Authority	users:write
PATCH	/users/{id}	Change role, jurisdiction or status; written reason required	Operator Portal, Authority	users:write
GET	/roles	The 15 roles and the permissions attached to each	Consoles	users:read

Method	Path	Purpose	Called by	Scope
POST	/ekyc/sessions	Open an eKYC session: ID OCR, liveness, face match against the registry	Citizen App, Driver App, Licensing	ekyc:write
GET	/ekyc/sessions/{id}	Poll the result; the score is advisory, a human decides	Citizen App, Driver App, Licensing	ekyc:read

5.2 Registry: operators, vehicles, drivers

Method	Path	Purpose	Called by	Scope
GET	/operators	Licensed bus companies, filterable by region, status and line held	Authority, Licensing, Citizen App	registry:read
GET	/operators/{id}	One company: depots, bank account, licence state, rating	All consoles	registry:read
PATCH	/operators/{id}	Update company profile; a settlement-account change needs two approvals	Operator Portal	operator:write
GET	/operators/{id}/lines	Lines this company is licensed to run, with permit references	Operator Portal, Authority	registry:read
GET	/vehicles	Fleet register; filter by operator, line, class, device and document status	Operator Portal, Authority	registry:read
POST	/vehicles	Register a vehicle; creates the licence application in the same call	Operator Portal	fleet:write
GET	/vehicles/{id}	One vehicle: seat layout, devices, documents, odometer, history	Operator Portal, Authority, Driver App	registry:read
PATCH	/vehicles/{id}	Change assignment, status or seat layout	Operator Portal	fleet:write
POST	/vehicles/{id}/block	Take a vehicle out of service; automatic on a critical inspection failure	Operator Portal, Authority	fleet:write
GET	/vehicles/{id}/documents	Licence, roadworthiness, insurance with expiry and OCR provenance	Operator Portal, Licensing	registry:read
GET	/drivers	Driver register with licence, medical, training and violation state	Operator Portal, Authority	registry:read
POST	/drivers/invite	Invite a driver by phone; they activate the app from the SMS	Operator Portal	drivers:write
POST	/drivers/self-register	Driver-initiated sign-up; the chosen operator must accept	Driver App	self:write

Method	Path	Purpose	Called by	Scope
GET	/drivers/{id}	One driver: documents, rating, hours this week, assigned vehicle	Operator Portal, Driver App	registry:read
PATCH	/drivers/{id}	Assign a vehicle, change status, record a violation	Operator Portal	drivers:write
GET	/drivers/{id}/hours	Hours of service against the fatigue rule, rolling 7 days	Driver App, Operator Portal	registry:read
GET	/terminals	Terminal register with geofence, bays, facilities and hours	All channels	—
GET	/terminals/{id}	One terminal, including operators serving it	All channels	—
PATCH	/terminals/{id}	Update facilities, hours, contact or geofence	Authority Console	terminal:write

5.3 Licensing and permits

Method	Path	Purpose	Called by	Scope
GET	/licensing/types	The six licence types with validity, fee basis and document checklist	Licensing Portal	—
POST	/licensing/applications	File an application; the fee is invoiced from the versioned schedule	Licensing Portal	licensing:apply
GET	/licensing/applications	Work queue, filterable by state, type, SLA breach and assignee	Licensing Portal, Authority	licensing:read
GET	/licensing/applications/{id}	One application with documents, checklist, timeline and invoice	Licensing Portal	licensing:read
POST	/licensing/applications/{id}/documents	Attach from the document vault; nothing is uploaded twice	Licensing Portal	licensing:apply
POST	/licensing/applications/{id}/verify	Officer records a pass or fail against each checklist item	Licensing Portal	licensing:verify

Method	Path	Purpose	Called by	Scope
POST	/licensing/applications/{id}/inspection	Book a ramp check and assign an inspector	Licensing Portal	licensing:verify
POST	/licensing/applications/{id}/approve	Approve and issue the e-permit; blocked for the verifying officer	Licensing Portal	licensing:approve
POST	/licensing/applications/{id}/reject	Reject with a written reason; the applicant may appeal	Licensing Portal	licensing:approve
POST	/licensing/applications/{id}/revision	Send back for correction with the list of defects	Licensing Portal	licensing:verify
GET	/permits	Permit register: terms, validity, holder, change history	All consoles	permits:read
GET	/permits/{id}	One permit with its versions and the reason for each change	All consoles	permits:read
PATCH	/permits/{id}	Amend terms; versioned, never retroactive to findings already issued	Authority Console	permits:write
POST	/permits/{id}/suspend	Suspend or revoke with a written reason and an effective date	Authority Console	permits:write
GET	/permits/{id}/certificate	Signed PDF with a verifiable QR	Licensing Portal, Operator Portal	permits:read
GET	/verify/{permit_id}	Public roadside verification of a permit QR — no login, no personal data	Public page, Citizen App	—
GET	/licensing/fees	Fee schedule with its version and effective dates	Licensing Portal	—
POST	/licensing/appeals	File an appeal; adjudicated by an officer who did not decide	Licensing Portal	licensing:apply

Method	Path	Purpose	Called by	Scope
GET	/documents/{id}	Fetch a vault document; virus-scanned, OCR-read, retention-tagged	Licensing, Operator Portal	documents:read
POST	/documents	Upload once, reuse across applications; returns extracted expiry dates	Licensing, Operator Portal	documents:write

5.4 Network, schedules and GTFS

Method	Path	Purpose	Called by	Scope
GET	/network/lines	Every published line with end points, class, holders and tracking state	All channels	—
GET	/network/lines/{id}	One line: stop sequence, shape, fare, operating hours, headway	All channels	—
POST	/network/lines	Register a new line; requires a route permit to exist first	Authority Console	network:write
PATCH	/network/lines/{id}	Change stops, shape or hours; checked against the permit in force	Authority Console	network:write
GET	/network/stops	Stop register with coordinates, facilities and the lines serving each	All channels	—
GET	/network/stops/nearby	Stops and terminals near a point, ordered by walking distance	Citizen App	—
GET	/network/shapes/{line_id}	The drawn geometry of a line for the map layer	All channels	—
GET	/schedules	Timetables by line and service day, frequency or fixed departures	All channels	—
POST	/schedules	Publish a timetable; validated against the permitted headway	Operator Portal	schedule:write
POST	/schedules/import	Bulk GTFS or CSV upload with a validation report before commit	Operator Portal, Authority	schedule:write
GET	/departures	Departures from a stop or terminal in a time window	Citizen App, board, Driver App	—
GET	/departures/{id}	One departure: vehicle, crew, bay, seats sold, live status	Citizen App, Operator Portal	—
POST	/departures/{id}/bay	Assign a bay; conflicts within 20 minutes are refused	Authority Console	terminal:write

Method	Path	Purpose	Called by	Scope
GET	/journeys/plan	Plan a trip from A to B: direct and one-transfer options, ranked	Citizen App	—
GET	/gtfs/static	GTFS-Static feed of the published network	Open data, third parties	—
GET	/gtfs/realtime/trip-updates	GTFS-Realtime TripUpdates protobuf	Open data, third parties	—
GET	/gtfs/realtime/vehicle-positions	GTFS-Realtime VehiclePositions protobuf	Open data, third parties	—
GET	/gtfs/realtime/alerts	GTFS-Realtime service alerts	Open data, third parties	—

5.5 Ticketing, seats and fares

Method	Path	Purpose	Called by	Scope
GET	/fares/quote	Price a journey: base fare, concession, YPS discount, promo	Citizen App, Boarding console	—
GET	/fares/bands	The floor and ceiling the authority publishes per line	All consoles, Citizen App	—
PATCH	/fares/bands/{line_id}	Set a band; written reason, effective date, never retroactive	Authority Console	fares:write
PUT	/fares/prices/{line_id}	Operator sets its selling price inside the band; 7-day notice	Operator Portal	fares:sell
GET	/inventory/{departure_id}/seats	Seat map with sold, held, priority and women-only seats	Citizen App, Operator Portal	—
POST	/inventory/{departure_id}/hold	Hold seats for ten minutes while the traveller pays	Citizen App	ticket:buy
DELETE	/inventory/holds/{id}	Release a hold early	Citizen App	ticket:buy
POST	/bookings	Create a booking: passengers, seats, payment intent. Idempotent.	Citizen App, Boarding console	ticket:buy
GET	/bookings/{id}	One booking with a ticket and a signed QR per seat	Citizen App	ticket:read
GET	/tickets	The caller's own tickets, upcoming and past	Citizen App	ticket:read
GET	/tickets/{id}	One ticket: QR, seat, boarding point, bay, calendar payload	Citizen App	ticket:read
POST	/tickets/{id}/check-in	Passenger self check-in inside the boarding window	Citizen App	ticket:read
POST	/tickets/{id}/cancel	Cancel and compute the refund against the policy in force	Citizen App	ticket:read

Method	Path	Purpose	Called by	Scope
POST	/tickets/{id}/reschedule	Move to another departure on the same line, fare difference applied	Citizen App	ticket:read
GET	/manifests/{departure_id}	Crew manifest: names and seats only, never passenger identity	Driver App, Operator Portal	manifest:read
GET	/refunds	Refund queue with policy outcome and dispute state	Operator Portal, Authority	refunds:read
POST	/refunds/{id}/decide	Approve or refuse a refund with a written reason	Operator Portal	refunds:write
GET	/promos	Promo codes and passes valid for the caller	Citizen App, Operator Portal	—
POST	/promos	Create a promo inside the fare band rules	Operator Portal	fares:sell

5.6 Payments and settlement

Method	Path	Purpose	Called by	Scope
GET	/payments/methods	Methods available for this booking, with the YPS discount applied	Citizen App	—
POST	/payments/intents	Create a payment intent for a booking. Idempotency-Key required.	Citizen App	ticket:buy
GET	/payments/intents/{id}	Poll the intent: pending, succeeded, failed, expired	Citizen App	ticket:buy
POST	/payments/intents/{id}/confirm	Confirm a card or wallet payment, including 3-D Secure return	Citizen App	ticket:buy
POST	/payments/webhooks/{provider}	Provider callback; signed, replay-protected, idempotent	KBZPay, Wave, AYA, CB, MPU	—
POST	/payments/yps/validate	Validate a YPS card tap or a YPS QR against the fare engine	Onboard POS, Driver App	pos:write
GET	/payments/yps/pos/{device_id}/health	POS readiness; checked before every daily departure	Operator Portal, Driver App	pos:read
GET	/wallet	Balance and transaction history for the signed-in passenger	Citizen App	wallet:read
POST	/wallet/topup	Top up from a payment method	Citizen App	wallet:write

Method	Path	Purpose	Called by	Scope
GET	/settlement/runs	T+1 settlement runs: gross, platform fee, levy, net per operator	Operator Portal, Authority	settlement:read
GET	/settlement/runs/{id}	One run with the ledger lines behind it	Operator Portal, Authority	settlement:read
GET	/settlement/exceptions	Three-way reconciliation breaks: provider, ledger, bank	Authority Console	settlement:read
POST	/settlement/exceptions/{id}/resolve	Resolve a break with a written reason before the day is signed off	Authority Console	settlement:write
GET	/revenue/levy	Government levy and licence fees by region, class and period	Authority Console	revenue:read
GET	/invoices	Invoices issued to an operator or an advertiser	Operator Portal, Authority	finance:read
POST	/invoices/{id}/pay	Pay an invoice; receipts are e-receipts with a verifiable number	Operator Portal, Licensing	finance:write

5.7 Tracking, telemetry and ETA

Method	Path	Purpose	Called by	Scope
POST	/telemetry/positions	Batched AVL ingest from the on-board unit: 1–10 s cadence	Telematics devices	telemetry:write
POST	/telemetry/events	Door, ignition, harsh braking, speeding, geofence, panic	Telematics devices	telemetry:write
GET	/tracking/vehicles	Live positions in a bounding box or on a line, jurisdiction-scoped	All consoles, Citizen App	tracking:read
GET	/tracking/vehicles/{id}	One vehicle: position, bearing, speed, occupancy, next stop	Consoles, Driver App	tracking:read
GET	/tracking/trips/{id}	Trip progress against the timetable, with the replay trail	Operator Portal, Authority	tracking:read
GET	/tracking/trips/{id}/replay	Historic breadcrumb trail for an investigation	Authority Console	tracking:replay
GET	/eta/stop/{stop_id}	Predicted arrivals at a stop, by line, with a confidence band	Citizen App, board	—
GET	/eta/trip/{trip_id}	Predicted arrival at each remaining stop of a running trip	Citizen App	—

Method	Path	Purpose	Called by	Scope
GET	/coverage	Coverage index: vehicles transmitting over vehicles expected	All consoles	tracking:read
GET	/devices	Telematics device register with last-seen and firmware	Operator Portal, Authority	devices:read
POST	/devices/{id}/command	Request a diagnostic or a firmware update from the unit	Operator Portal	devices:write
GET	/tracking/stream	Server-sent events for a console map; falls back to polling	Consoles	tracking:read

5.8 Boarding and validation

Method	Path	Purpose	Called by	Scope
POST	/boarding/scan	Validate a passenger QR at the door or the gate. Works offline, syncs later.	Operator Portal, gate console	boarding:write
POST	/boarding/cash	Record a cash passenger so the run reconciles	Operator Portal, gate console	boarding:write
GET	/boarding/runs/{departure_id}	Expected, boarded, cash and rejected counts for a run	Operator Portal, Driver App	boarding:read
GET	/boarding/keys	Public keys for offline QR verification; rotated, cached on device	Driver App, gate console	boarding:read
POST	/shifts/check-in	Start a shift: face match, bus QR, fitness declaration	Driver App	shift:write
POST	/shifts/{id}/inspection	Pre-departure vehicle inspection; a critical fail blocks the bus	Driver App	shift:write
POST	/shifts/{id}/end	End the shift; tracking stops at this moment	Driver App	shift:write
GET	/shifts/current	The caller's open shift, if any, with its assigned trips	Driver App	shift:read
GET	/gate/{terminal_id}/log	Gate events for the shift, feeding the crowd monitor	Authority Console	terminal:read

5.9 Safety, SOS and incidents

Method	Path	Purpose	Called by	Scope
POST	/sos	Raise an SOS from a passenger or a driver. Always P1.	Citizen App, Driver App	sos:write
GET	/sos/active	Open SOS board with map position and evidence	Authority Console, Operator Portal	sos:read

Method	Path	Purpose	Called by	Scope
POST	/sos/{id}/acknowledge	Acknowledge inside the 5-minute SLA; escalates after 2 minutes	Authority Console	sos:write
POST	/sos/{id}/dispatch	Dispatch police, medical or the operator's own response	Authority Console	sos:write
POST	/incidents	Report an incident with category, location and optional evidence	Citizen App, Driver App, Operator	incidents:write
GET	/incidents	Incident queue by priority, status, line and acknowledgement deadline	Consoles	incidents:read
GET	/incidents/{id}	One incident with its timeline and every state change	Consoles	incidents:read
POST	/incidents/{id}/forward	Forward to the operator or another authority with a reason	Authority Console	incidents:write
POST	/incidents/{id}/close	Close with a written resolution; cannot be undone	Authority Console	incidents:write
POST	/incidents/{id}/evidence	Attach photo, audio or the telemetry window around the event	All channels	incidents:write
GET	/safety/contacts	Emergency numbers and the trusted contacts the passenger set	Citizen App	self:read
POST	/trips/{id}/share	Share a live trip with a trusted contact by SMS link	Citizen App	ticket:read

5.10 Compliance and enforcement

Method	Path	Purpose	Called by	Scope
GET	/compliance/matrix	Per line: every measurable dimension against the threshold in force	Authority, Operator Portal	compliance:read
GET	/compliance/lines/{id}	One line with its measurement window, coverage and verdict	Authority, Operator Portal	compliance:read
POST	/compliance/findings	Issue a formal finding with the measurement behind it	Authority Console	compliance:write
GET	/compliance/findings	Findings by state: issued, operator responded, disputed, upheld, closed	Authority, Operator Portal	compliance:read

Method	Path	Purpose	Called by	Scope
POST	/findings/{id}/dispute	Operator disputes with evidence; a different officer rules	Operator Portal	compliance:respond
POST	/findings/{id}/rule	Uphold or set aside a dispute with a written reason	Authority Console	compliance:approve
POST	/enforcement/fines	Issue an e-fine; links to the police e-ticketing system	Authority Console	enforcement:write
GET	/enforcement/sanctions	Active sanctions against operators, vehicles and drivers	Authority, Operator Portal	enforcement:read
GET	/thresholds	Service standards, SLAs, coverage floors and k-anonymity, with version	All consoles	—
PATCH	/thresholds	Change a threshold; national policy role only, written reason, versioned	Authority Console	thresholds:write
GET	/audit	Append-only audit log, filterable by actor, object and period	Authority Console	audit:read
GET	/audit/{id}/chain	Verify the hash chain around one entry	Authority Console	audit:read

5.11 Content, notifications and advertising

Method	Path	Purpose	Called by	Scope
GET	/content/announcements	Published service alerts for a region, line or terminal	All channels	—
POST	/content/announcements	Draft an alert; published only after authority approval	Authority Console	content:write
POST	/content/announcements/{id}/publish	Publish to app, board and push in one action	Authority Console	content:approve
GET	/content/articles	Help centre and legal pages; the same corpus the chatbot may quote	Citizen App, Driver App	—
GET	/content/home-layout	The order of blocks on the Citizen App home screen	Citizen App	—
PUT	/content/home-layout	Reorder or hide blocks; search and safety cannot be removed	Authority Console	content:write

Method	Path	Purpose	Called by	Scope
POST	/notifications/send	Send from a versioned template on a chosen channel	Authority, Operator Portal	notify:write
GET	/notifications	The caller's inbox	All channels	self:read
POST	/notifications/{id}/read	Mark read; drivers must acknowledge operational messages	All channels	self:write
GET	/notifications/templates	Templates in Myanmar and English with their versions	Authority Console	notify:read
GET	/notifications/delivery	Delivery receipts, retries and push-to-SMS fallback	Authority Console	notify:read
GET	/ads/inventory	Ad slots with location, format, rate and sold percentage	Authority, Operator Portal	ads:read
POST	/ads/campaigns	Book a campaign with contextual targeting only	Operator Portal, advertisers	ads:write
POST	/ads/campaigns/{id}/approve	Creative approval against the advertising policy	Authority Console	ads:approve
GET	/ads/delivery	Aggregate delivery: impressions, clicks, spend. No passenger identity.	Authority, advertisers	ads:read
GET	/helpdesk/tickets	Complaints from app, chatbot and web in one queue with SLA	Authority, Operator Portal	helpdesk:read
POST	/helpdesk/tickets/{id}/reply	Reply and resolve; published replies are visible to the passenger	Authority, Operator Portal	helpdesk:write

5.12 Analytics, reporting and open data

Method	Path	Purpose	Called by	Scope
GET	/analytics/kpis	The KPI set for a jurisdiction and period, with coverage labels	All consoles	analytics:read
GET	/analytics/demand	Origin–destination pairs and unserved searches, k-anonymised	Authority Console	analytics:read
GET	/analytics/corridors	Speed and occupancy by corridor and two-hour window	Authority Console	analytics:read
GET	/analytics/operators	Operator league table on published measures only	Authority Console	analytics:read

Method	Path	Purpose	Called by	Scope
POST	/reports	Build a report from the catalogue; line-level export needs a reason	Authority, Operator Portal	reports:write
GET	/reports/{id}	Poll and download; the data-limitations block travels with the file	Authority, Operator Portal	reports:read
GET	/reports/catalogue	Operational and ministry report definitions	Consoles	reports:read
GET	/opendata/ridership	Aggregate ridership, open licence, k-anonymity enforced	Public	—
GET	/opendata/punctuality	Published punctuality by line and month	Public	—
GET	/planning/scenarios	What-if: extend a line, add a corridor, change a headway	Authority Console	planning:read
POST	/planning/scenarios	Run a scenario against the demand model	Authority Console	planning:write

6. API scope per portal

The same catalogue, cut by channel. A team building one portal can take its column and ignore the rest. **W** means the channel writes as well as reads.

Domain	Citizen	Driver	Operator	Licensing	Authority	Board
Identity and access	read + W	read + W	read + W	read + W	read + W	token only
Registry (operators, vehicles, drivers)	read (public fields)	read (own bus)	read + W	read + W	read + W	—
Licensing and permits	verify QR only	—	read + W (own)	read + W	read + W	—
Network, schedules, GTFS	read	read (own line)	read + W	read	read + W	read
Ticketing, seats, fares	read + W	manifest read	read + W	—	read (bands) + W	—
Payments and settlement	read + W	—	read + W	invoice pay	read + W (oversight)	—
Tracking, telemetry, ETA	read (own trip, nearby)	read + W (own shift)	read + W (own fleet)	—	read (jurisdiction)	read (one terminal)
Boarding and validation	check-in only	read + W	read + W	—	read + W (gate)	—
Safety, SOS, incidents	read + W	read + W	read + W	—	read + W	—
Compliance and enforcement	—	—	read + respond	—	read + W	—
Content, notifications, ads	read	read + ack	read + W (ads)	—	read + W	read (alerts)
Analytics and reporting	—	own stats	read + W (own)	read (own)	read + W	—

The operator API is a product, not an afterthought. A bus company that already runs its own reservation or fleet system must be able to keep it in step: booking, check-in and SOS are exposed as OpenAPI 3 endpoints with client-credentials tokens scoped to that operator, plus webhooks for `booking.created`, `booking.cancelled`, `vehicle.blocked` and `sos.raised`. Rate limiting is applied at the gateway, not inside the integration.

7. Front-end specification

7.1 Screen inventory

126 screens across six deployments, built from one codebase. A single-portal build is produced by setting VITE_PORTAL, so each channel gets its own URL while the routing, components and data contracts stay shared.

Channel	Screens	Shell	Route prefix	Deployment
Citizen App	26	Phone-width column, 3 bottom tabs	/citizen	yangon-citizen
Driver App	14	Phone-width column, 4 bottom tabs, dark app bar	/driver	yangon-driver
Bus Operator Portal	26	Desktop sidebar, 5 nav groups	/operator	yangon-operator
Licensing Portal	15	Desktop sidebar, applicant / officer switch	/licensing	yangon-licensing
Authority Console	42	Desktop sidebar, 6 nav groups, jurisdiction picker	/authority	yangon-authority
Public pages	3	Full-bleed, no sign-in	/board/:id, /verify/:id, /blueprint	yangon-terminal

7.2 Navigation map per portal

Citizen App — the buying journey

Three bottom tabs only: Home, Tickets, Account. Everything else is reached from a card or a search.

Welcome (phone → OTP) → Home
 Home → Search destination → Journey preview (leave now / later) → Results
 Results → Line detail → **Departure picker** (date strip, sort by time / price / seats)
 Departure picker → **Seat map** (10-min hold) → **Passenger details** → **Review & pay** → **E-ticket**
 Home → Nearby stop / terminal → Terminal detail (departures, routes from here, arrivals)
 E-ticket → Check-in → **Live trip** (opens 30 min before departure) → Share trip · SOS
 Tickets → Upcoming / History → E-ticket
 Account → Profile · My tickets · Favourites · Wallet · Payment methods · SOS · Report an incident · My reports · Verify a bus licence · Ask YanGo · Notifications · Language · Accessibility · Privacy

Driver App — one shift, one bus

Welcome → invited by company (SMS + OTP) or self-register → Onboarding → Home
 Home → **Shift check-in** (1 face match → 2 bus QR → 3 fitness) → On duty
 On duty → Trips → Trip detail → Manifest · Inspection · Fatigue & safety
 Any state → **SOS** (always reachable, not only on duty)
 Home → End shift (odometer, fuel, defects) → Off duty, tracking stops
 Tabs: Home · Trips · Inbox · Profile

Bus Operator Portal — running the company

Overview Dashboard · Live operations
Fleet & people Fleet → Vehicle detail · Maintenance · Drivers · Roster
Network & sales Routes · Schedules · Fares & promos · Seat inventory · Bookings & manifest ·

Boarding & check-in (the conductor scans here, not the driver) · Refunds

Safety & compliance SOS & incidents · Compliance → dispute a finding

Commercial Finance & settlement · Ads & promotions · Reviews · Reports

Administration Company profile · Users & roles · Licences · API & webhooks · Audit log

Licensing Portal — two seats at one desk

Applicant Home → New application wizard (1 licence type → 2 details → 3 documents → 4 review)

→ Invoice → Pay → Tracker → My licences → File an appeal

Officer Work queue → Application detail → Verification checklist → Inspection → Send to approver

→ **Approver** (must be a different officer) → Approve and issue e-permit / Reject / Request revision

Administration Fee schedule · Licence register · Document vault · Licensing reports

Authority Console — the regulator's desk

Operations Command center · Incidents · SOS live board · Compliance · Enforcement

Insight Analytics · Planning · Reports · National dashboard · Regional dashboard · AI & automation

Licensing & money Licensing overview · Permit register · Regulated revenue · Settlement oversight

Registries Operators, fleet & drivers · **YBS line register** · Fare bands

Terminals Terminal overview · Bay allocation · Departure board · Gate check-in · Crowd monitor

Public information Announcements · Knowledge base · Home layout · Templates · Broadcasts · Delivery · Helpdesk

Advertising Inventory · Campaigns · Advertisers · Billing · Policy

Platform admin Users · Roles & permissions · Thresholds · Integrations · Audit log

7.3 Screen-to-endpoint binding

The screens that carry the most traffic, and what each one calls. Everything else follows the same pattern: one read on mount, one write on the primary action.

Screen	On mount	On action
Citizen · Home	GET /network/stops/nearby, GET /eta/stop/{id}, GET /tracking/vehicles, GET /content/announcements, GET /tickets?status=active	navigate only
Citizen · Departure picker	GET /departures?line&date, GET /fares/quote	select → seat map
Citizen · Seat map	GET /inventory/{dep}/seats	POST /inventory/{dep}/hold
Citizen · Review & pay	GET /payments/methods, GET /fares/quote	POST /bookings → POST /payments/intents → poll
Citizen · E-ticket	GET /tickets/{id}	POST /tickets/{id}/check-in
Citizen · Live trip	GET /eta/trip/{id}, GET /tracking/vehicles/{id} (SSE)	POST /trips/{id}/share, POST /sos
Citizen · Terminal detail	GET /terminals/{id}, GET /departures?terminal={id}, GET /network/lines?terminal={id}	navigate to booking
Driver · Home	GET /shifts/current, GET /drivers/{me}/hours, GET /schedules?driver=me	navigate
Driver · Shift check-in	GET /vehicles/{id}	POST /ekyc/sessions → POST /shifts/check-in

Screen	On mount	On action
Operator · Dashboard	GET /analytics/kpis , GET /tracking/vehicles?operator=me , GET /sos/active , GET /compliance/findings , GET /permits?expiring	navigate
Operator · Boarding	GET /departures?operator=me&next , GET /boarding/runs/{dep} , GET /boarding/keys	POST /boarding/scan , POST /boarding/cash
Authority · Command center	GET /analytics/kpis , GET /tracking/vehicles , GET /incidents?open , GET /sos/active , GET /coverage	POST /sos/{id}/acknowledge
Authority · Fare bands	GET /fares/bands , GET /network/lines	PATCH /fares/bands/{line} + reason
Licensing · Application detail	GET /licensing/applications/{id} , GET /documents	POST .../verify , POST .../approve
Public · Departure board	GET /departures?terminal={id} , GET /content/announcements	auto-refresh 20 s

7.4 State, caching and offline behaviour

Concern	Rule
Server state	One query client with stale-while-revalidate. Reference data (lines, stops, fares) cached 15 minutes; live data (positions, ETA) never cached.
Live updates	Server-sent events on console maps and the live trip screen; polling at 10 s is the fallback when SSE is blocked.
Offline — Citizen	Tickets and their signed QR are written to IndexedDB. A ticket scans at the door with no connection at all.
Offline — Driver	Shift events, inspections, scans and SOS are queued locally and replayed in order when signal returns. An SOS raised offline is sent by SMS as well.
Optimistic UI	Only for actions that cannot fail server-side (marking a notification read). Never for payment, approval or issuance.
Error surface	Every failed write shows what failed, what the user can do, and keeps their input. No silent retries on a payment.
Language	Passenger and driver language live in the session; console language lives on the workstation. They never leak into each other.
Accessibility	WCAG 2.1 AA: 44 px touch targets, visible focus, labelled controls, 13 px minimum body text, full Myanmar Unicode.

8. Tracking and maps engine

This is the part that decides whether the platform is credible. A regulator that publishes a punctuality figure has to be able to defend how it was measured, and a passenger who is told the bus is four minutes away has to find it there.

8.1 Ingest

Item	Specification
Device	On-board unit with GNSS (GPS + GLONASS/BeiDou), GSM/LTE, internal battery, tamper flag. IMEI registered against the vehicle in the fleet register — an unregistered IMEI is rejected at the edge.
Protocol	MQTT over TLS preferred; HTTPS batch POST as fallback for units that cannot hold a session. Both carry the same payload schema.
Cadence	5 s while moving, 30 s while stationary with the ignition on, 5 min with the ignition off. Position bursts to 1 s for 60 s after an SOS or a harsh-event trigger.
Payload	device_id, ts, lat, lng, speed, heading, hdop, satellites, ignition, door_state, odometer, occupancy (where an APC is fitted), event_code, queue_depth.
Buffering	The unit stores at least 24 h of positions and replays them in order when signal returns. Replayed points carry their original timestamp and are flagged <code>late</code> .
Volume (pilot)	~420 instrumented vehicles × 12 points/min ≈ 5 000 points/min at peak, ≈ 7.3 M points/day. Sized for 2 000 vehicles from day one.
Pipeline	Gateway → message broker (Kafka or equivalent) → stream processor → hot store (time-series, 30 days) and cold store (object storage, Parquet, 3 years).

8.2 Map matching and trip inference

- **Map matching** — an HMM matcher (OSRM or Valhalla against an OpenStreetMap extract of Myanmar) snaps each point to the road graph and then to the line's shape. Points more than 80 m from any candidate shape mark the vehicle *deviating*.
- **Trip inference** — a run is opened when the vehicle leaves the origin geofence with an open shift, and closed at the destination geofence or after 20 minutes of no movement at a terminal.
- **Stop arrival** — recorded on geofence entry (40 m radius) plus a door-open event where the unit reports one; dwell is the interval to geofence exit.
- **Headway** — measured between consecutive arrivals of the same line at the same stop, not from the timetable. Bunching is two buses within 40 % of the planned headway.

8.3 ETA

Two models, one contract. The API always returns a point estimate **and** a confidence band; a client never shows a bare number it cannot defend.

- **Baseline** — remaining shape distance divided by a segment speed profile learned per line, per direction, per two-hour window, per day type. Available from day one, no training needed.
- **Learned** — gradient-boosted regression over the baseline plus live speed, dwell history, rainfall and public-holiday flags. Ships once 8 weeks of matched history exists.
- **Fallback** — if coverage on a line drops below the floor, the API returns the scheduled time and labels it `scheduled`, never a prediction dressed as live.
- **Target** — median absolute error under 120 s at 15 minutes out, measured weekly and published on the AI page.

8.4 Maps, tiles and geocoding

Layer	Choice	Why
Base tiles	Self-hosted vector tiles from an OpenStreetMap Myanmar extract, served with MapLibre GL	No per-view licence fee, no third-party dependency for a government service, and the Burmese labels in OSM are good in Yangon
Fallback tiles	OSM raster tiles	Used in the prototype and as a degraded mode
Routing / matching	OSRM or Valhalla, self-hosted	Map matching and walking directions without sending passenger positions to a third party
Geocoding	Photon or Nominatim over the same extract, seeded with the stop and terminal register	Place search has to find Burmese stop names first, streets second
Offline	Stop and line geometry bundled with the app; the map degrades to a stop list without tiles	Rural coverage and cheap devices

Do not put passenger coordinates through a commercial maps API. Nearby-stop search, journey planning and live tracking all run on self-hosted engines. The only thing that leaves the platform is an anonymous tile request.

8.5 Coverage index

Everything above is worthless if the buses are not transmitting, so coverage is a first-class figure: vehicles transmitting in the last 10 minutes over vehicles expected in service. It is attached to every derived measure. Above 80 % figures are published in full; between 50 % and 80 % they are labelled indicative; below 50 % the line is **not measurable** — never *failed*.

9. AI scope

AI here is not a chatbot bolted to a help page. It is 13 bounded use cases, each with a named input, a named output and a named screen it appears on, plus three assistants. Every output is advisory: a named officer still makes each regulatory decision.

9.1 Use cases

ID	Use case	Pri	Phase	Reads	Produces	Appears in
AI-01	ETA prediction	M	1	AVL position history (10 s), time of day, day type, traffic speed, weather	Arrival time per stop with a confidence band	Citizen arrivals list, trip tracker, terminal departure board, GTFS-RT feed
AI-02	Document AI (OCR + tamper check)	M	1	Uploaded IDs, driving licences, roadworthiness and insurance certificates	Extracted fields, expiry date, tamper score	Document vault, verification checklist, expiry reminders
AI-03	Face match & liveness	M	1	Selfie frame, ID photo	Match score, liveness result	Shift check-in gate, driver registration, applicant KYC
AI-04	Driving-behaviour scoring	S	2	Speed profile, harsh braking and cornering events, route deviation, driving hours	Driver risk score and event alerts	Driver fatigue screen, operator driver list, authority safety view
AI-05	Anomaly & fraud detection	S	2	Payments, refunds, bookings, applications, GPS traces	Flags: duplicate identities, refund abuse, ghost trips, GPS spoofing	Reconciliation exception queue, licensing duplicate check, settlement holds
AI-06	Demand forecasting	S	2	Ticket sales, journey searches, calendar, festivals, weather	Forecast passengers per route and hour, extra-trip suggestions	Planning forecast overlay, operator schedule editor
AI-07	Schedule & roster optimisation	C	3	Demand forecast, permit terms, fleet availability, hours-of-service rules	Proposed timetable and roster that stay inside the permit	Timetable editor, roster conflict checker
AI-08	Incident triage	S	2	Report text, category, location, audio, recent incident history	Suggested priority, category and duplicate cluster	Incident drawer (advisory panel), SLA routing
AI-09	Dashcam video analytics	C	3	MDVR clips attached to an incident	Distraction, phone use, overcrowding indicators	Incident evidence panel
AI-10	Occupancy estimation	S	2	Automatic passenger counters, gate check-ins, camera counts	Occupancy band: empty / some seats / full	Arrivals list, crowd monitor, occupancy analytics
AI-11	Predictive maintenance	C	3	Defect history, CAN bus data, mileage, inspection results	Failure risk per vehicle and component	Maintenance schedule, work-order prioritisation
AI-12	Ad creative moderation	S	2	Creative images and copy	Policy flags against prohibited categories	Approval queue advisory panel
AI-13	Report narrative drafting	C	3	k-anonymous aggregate marts for the selected period	Draft commentary for an officer to edit	Report sections, weekly brief

9.2 Assistants

Assistant	Users	May do	Reads	Produces
Citizen assistant — “Ask YanGo”	Passengers	['Plan a journey in Myanmar or English, including by voice', 'Next bus, fares, terminal facilities and accessibility', 'Status of the traveller's own ticket, refund policy, lost & found', 'Open a complaint or surface the SOS button']	Journey planner, GTFS-RT, fare API, own-ticket API (authenticated), CMS / FAQ, terminal registry	Answers with a cited source, deep links (pre-filled booking, SOS, complaint), handover to a human agent with the transcript
Operator assistant	PO staff	['Explain the permit terms that apply to a line', 'Check a draft timetable against minimum headway and vehicles held', 'Summarise incidents and draft dispute text', 'Explain a settlement line and find expiring licences']	Operator-scoped APIs, permit terms, regulation library	Answers with citations; drafts the user edits and submits themselves
Officer assistant	Authority officers	['Summarise network status for a shift handover', 'Draft the weekly brief from aggregate marts', 'Explain how a figure was produced and what qualifies it', 'Pre-screen an application for completeness and search the regulations']	k-anonymous marts, licensing records, regulation library, published reports	Summaries; draft findings and letters that the officer signs

The citizen assistant answers from approved content and the caller's own booking data only. It may not quote an unapproved source, and it hands over to a human with the transcript when it cannot answer. The officer assistant drafts; it never decides.

9.3 Guardrails — applied to every assistant on every turn

- Only the user's own session data enters a prompt; personal data is redacted in logs.
- Answers cite the source article or dataset; legal and medical advice is refused.
- Safety keywords (accident, harassment, fire) instantly surface the SOS button.
- Conversation logs are kept 90 days and used for training only in anonymised form with consent.
- Containment rate, CSAT and escalation rate are reported monthly, with a hallucination audit.
- No AI output is a decision. Every regulatory action still needs a named officer and a written reason.

Enforced at the gateway, not in the prompt. Scope, rate and data-access limits are applied by the service in front of the model. A prompt that asks the assistant to ignore its instructions still cannot reach a row the caller has no right to read.

9.4 Model quality

Measure	Now	Target
Chatbot containment	63 %	60 %
Chatbot CSAT	4.2 / 5	4 / 5
Escalation to a human	18 %	25 %
ETA within ± 2 min	81 %	80 %
Document OCR accuracy	96.4 %	95 %
Face match false reject	1.8 %	3 %

9.5 Serving

- **Demand and ETA models** — trained offline, versioned, served behind a feature store. A model change has a known blast radius because every consumer screen is registered against it.
- **Assistants** — a hosted LLM behind the platform's own gateway, with retrieval restricted to the approved knowledge base and the caller's own records. No passenger identity ever enters a prompt.
- **Evaluation** — containment, handover rate and answer accuracy are measured weekly against a held-out set and published on the AI & automation page.
- **Rollback** — every model is pinned by version per environment; a bad release is rolled back without a code deploy.

10. External integrations

15 external engines. Each one has a named fallback, because a government service cannot stop when a third party does.

ID	Engine	Counterparty	Exchange	Pri	Fallback
INT-01	National ID / eKYC	National ID registry; eKYC vendor	ID number, photo → match result	M	Manual verification against the uploaded document
INT-02	Driving licence	Road transport administration licence database	Licence number → class, validity, suspension	M	Officer checks the scanned licence and records the check
INT-03	Vehicle registry	Vehicle registration and roadworthiness records	Plate / chassis → owner, type, test validity	M	Roadworthiness certificate upload plus ramp check
INT-04	Company & tax registry	Company registry (DICA MyCO), tax authority	Registration number → status, directors	M	Certified company extract upload
INT-05	Payment providers	KBZPay, Wave Money, AYA Pay, CB Pay, MPU, Visa/Mastercard, banks	Charge, callback, refund, settlement file	M	Counter payment with numbered receipt
INT-06	Treasury	Government revenue account	Fee and fine postings, reconciliation	M	Daily manual posting from the reconciliation report
INT-07	SMS / OTP	Mobile operators' SMS gateways	OTP, alerts	M	Push only, with in-app inbox
INT-08	Push & messaging	FCM, APNs, Viber Business, WhatsApp Business, SMTP	Notifications	M	SMS
INT-09	Telematics	GPS trackers, MDVR dashcams, APC sensors, panic buttons (GT06 / JT808 over MQTT or TCP)	Position every ≤ 10 s, alarms, video clips, counts	M	Driver-phone GPS as an interim source
INT-10	Maps & geocoding	OpenStreetMap tiles, self-hosted geocoder and routing	Tiles, addresses, routes	M	Schematic map without tiles (lite mode)
INT-11	Emergency dispatch	Police, ambulance, fire, traffic control	SOS with location, vehicle, driver	M	Duty officer telephones the control room; the call is logged
INT-12	Insurance	Insurers	Policy verification, accident claims	S	Policy certificate in the document vault
INT-13	Enforcement / e-fine	Traffic police e-ticketing, roadside inspection devices	Violations, fines	S	e-Fine issued in-platform and re-keyed by the enforcement desk
INT-14	Open data & partners	GTFS / GTFS-RT public feed, map platforms, ticket resellers	Schedules, real time, bookings	S	Scheduled file export

ID	Engine	Counterparty	Exchange	Pri	Fallback
INT-15	AI / LLM gateway	In-country LLM, Myanmar speech-to-text	Prompts, answers	S	Chatbot disabled; FAQ and human helpdesk remain

11. Non-functional requirements

Area	Requirement
Performance	Search results under 2 s at p95. QR validation under 300 ms at p95, offline path under 100 ms. Console list views under 1.5 s for 10 000 rows. Map first paint under 2.5 s on a mid-range Android phone over 3G.
Availability	99.9 % monthly for ticketing, tracking and SOS. 99.5 % for consoles. Planned maintenance outside 05:00–23:00 Yangon time.
Scale	Designed for 1.5 M passenger journeys a day, 2 000 instrumented vehicles, 135 lines and 12 operators in Yangon Region, with headroom for three more regions without re-architecture.
Resilience	No single point of failure in ingest or payment. A provider outage degrades to cash and to queued scans, never to a stopped bus.
Security	TLS 1.3, mandatory 2FA for every console user, least-privilege scopes, secrets in a managed vault, quarterly penetration test, signed and versioned QR keys with rotation.
Privacy	Passenger identity never reaches an authority dashboard. Aggregates covering fewer than 20 journeys are not produced, and the rule is enforced in the data layer. Location history is retained 90 days for passengers, 3 years for regulated vehicle telemetry.
Data residency	Primary storage in Myanmar where the law requires it; backups encrypted, key custody with the authority.
Auditability	Append-only, hash-chained log of every regulatory action with actor, object, threshold version and written reason. Exports carry their data-limitations block.
Accessibility	WCAG 2.1 AA on every citizen and driver screen; Myanmar Unicode throughout; 13 px minimum body text; 44 px touch targets.
Localisation	Myanmar and English across all six channels. Passenger language and console language are separate settings.
Observability	Structured logs with a request id that survives every hop, RED metrics per endpoint, distributed tracing, and an SLO dashboard the authority can see.
Open data	GTFS-Static and GTFS-Realtime published under an open licence, plus aggregate ridership and punctuality.

12. Delivery phases and open questions

12.1 Phases

Phase	Scope	Proves
1 — Foundations	Identity, registries, licensing workflow, permits, network master, GTFS-Static, Authority Console read-only	One operator can be licensed end to end and appear on a map
2 — Movement	Telemetry ingest, map matching, live tracking, baseline ETA, coverage index, departure boards, Driver App shifts	The regulator can see the network move and defend the numbers
3 — Money	Ticketing, seat inventory, fares and bands, payments including YPS, boarding validation, settlement T+1	A passenger buys, boards and the operator is paid the next day
4 — Oversight	Compliance matrix, findings and disputes, enforcement, SOS board, analytics, reporting, open data	Regulation is evidence-based and appealable
5 — Intelligence	Learned ETA, demand forecasting, planning scenarios, assistants, anomaly detection	The network is planned from what actually happened

12.2 Open questions for YRTC

1. **Fare authority.** The cash fare is reported at K400 and YPS is cheaper. Confirm the current figure, the exact YPS differential, and who may change a band — region or ministry.
2. **Line register.** The public directory lists roughly 135 lines with branches under one number (7, 31, 34, 64, 89, 111). Confirm the canonical list and how a branch is identified, because it decides the primary key of the whole network model.
3. **Operator mapping.** Directory listings and the YRTC route-map announcement disagree on which company holds which line. The registry needs one authoritative source.
4. **YPS integration.** Confirm the interface to the YPS card scheme and the MPyps+ QR — whether the platform validates against YPS or YPS settles to the platform.
5. **Telematics estate.** How many vehicles already carry a unit, of what make, and who owns the data today. This sets the real coverage index on day one.
6. **Data residency.** Confirm which datasets must remain inside Myanmar and who holds the backup encryption keys.

A note on the data in the working build. Line numbers, end points and operator names come from the public Yangon bus directory and the YRTC route-map announcement. Coordinates for stops not yet surveyed are approximate, and contact and banking details are synthetic. All of it is to be reconciled with YRTC before any of it is treated as a register.